

## Workshop 6

1. Sketch the graph of the function  $f(x) = \ln |\sin(x)|$ .

2. (Final B, Spring 2024) Find the global extrema of

$$f(x) = x^4 - 64 \ln\left(\frac{x}{6}\right), \quad [1, 6]$$

Hint:  $\ln(2) = 0.693\dots$ ,  $\ln(3) = 1.098\dots$

3. (Final A, 2021) Let  $f[a, b] \rightarrow \mathbb{R}$  be a continuous function which is twice differentiable on  $(a, b)$ . Assume that there exists  $c \in (a, b)$ , such that  $f(a) = f(c) = f(b)$ . Prove that there exists at least one point  $\alpha \in (a, b)$ , such that  $f'(\alpha) = 0$ .

4. (Final A, September 2022) Solve the following limit

$$\lim_{x \rightarrow 0} \frac{x2^x}{2^x - 1},$$

5. (Final Sept 2022) Consider the following function

$$f(x) = \frac{x^2 - 49}{x^2 + 5x - 14}.$$

Find the following

- (a) The domain and symmetries,
  - (b) Asymptotes
  - (c) Critical points,
  - (d) Local maximum and minimum,
  - (e) Increasing/decreasing intervals,
  - (f) Inflexion points,
  - (g) Concavity intervals,
  - (h) Sketch the graph.
6. (Final A, 2021) Consider  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = e^x(ax^2 + b)$ , where  $a, b \in \mathbb{R}$  are real constants.
- (a) Under which conditions does  $f$  have at least two critical points?
  - (b) Assume  $a = 4$  and  $b = 3$ . Find the increasing and decreasing intervals of  $f$ , as well as the local extrema points.